

Problem 1: Let $V = P_2(\mathbb{R})$ and $T \in L(P_2(\mathbb{R}))$. Then check whether the operator T is diagonalizable when

(i) $T(p(x)) := p'(x) + p''(x)$

(ii) $T(ax^2 + bx + c) := cx^2 + bx + a$

(iii) $T(p(x)) := p(0) + p(1)(x + x^2)$.

Solution: (i) $T(p(x)) = p'(x) + p''(x)$.

Let $\beta = (1, x, x^2)$ be an ordered basis of $P_2(\mathbb{R})$.

Then let us calculate $[T]_{\beta}^{\beta}$

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2x + 2$$

Then

$$[T]_{\beta}^{\beta} = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

The characteristic poly of $[T]_{\beta}^{\beta}$ is

$$\det([T]_{\beta}^{\beta} - \lambda I_3) = \det \begin{pmatrix} -\lambda & 1 & 2 \\ 0 & -\lambda & 2 \\ 0 & 0 & -\lambda \end{pmatrix}$$

$$= (-\lambda)^3$$

Hence the eigenvalue of T is 0.

Since T is not the zero linear transformation

$$E_{\lambda_0} \subsetneq \mathcal{P}_2(\mathbb{R}) \text{ where } \lambda_0 = 0$$

Hence $\nexists$ a basis consisting of eigenvectors of T .

Hence T is not diagonalizable.

$$(ii) \quad T(ax^2+bx+c) = cx^2+bx+a$$

$$T(1) = x^2, \quad T(x) = x, \quad T(x^2) = 1.$$

Hence

$$[T]_{\beta}^{\beta} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

char. poly

$$\det([T]_{\beta}^{\beta} - \lambda I_3)$$

$$= \det \begin{pmatrix} -\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{pmatrix} = \begin{matrix} \lambda^2(1-\lambda) - (1-\lambda) \\ (\lambda^2-1)(1-\lambda) = -(\lambda+1)(\lambda-1)^2 \end{matrix}$$

Hence the eigenvalues of T are $\lambda_1 = 1$ $\lambda_2 = -1$

$$\text{If } ax^2+bx+c = T(ax^2+bx+c) = cx^2+bx+a$$

$$\Rightarrow a=c \quad \text{Hence if } ax^2+bx+c \text{ is an eigenvector}$$

$$\text{then } a=c. \text{ If } a=c, T(ax^2+bx+a) = ax^2+bx+a$$

$$\Rightarrow E_{\lambda_1} = \{ax^2+bx+c : a=c\} = \{a(x^2+1)+bx : a, b \in \mathbb{R}\}$$

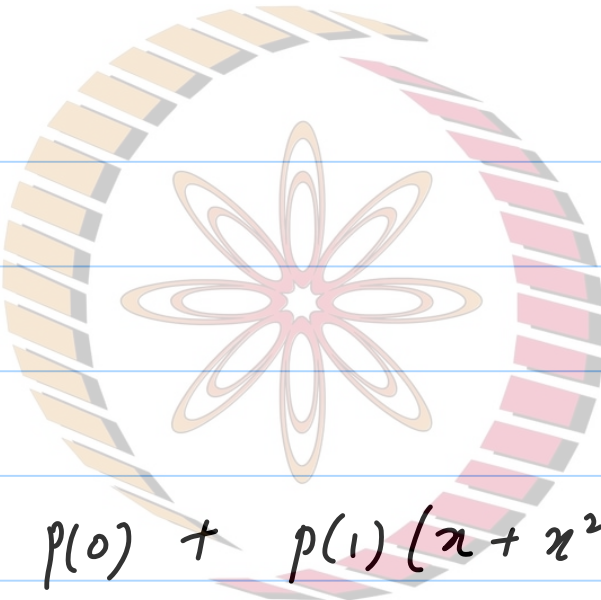
$$= \text{span} \{ (1+x^2), x \}.$$

$$\text{If } -(ax^2+bx+c) = T(ax^2+bx+c) = cx^2+bx+a$$

$$\Rightarrow a=-c, \quad 2b=0 \quad E_{\lambda_2} = \{a(x^2-1) : a \in \mathbb{R}\}.$$

Hence $\beta = \{ (1+x^2), x, x^2-1 \}$ are eigenvectors of T .

$\Rightarrow \beta$ is a basis of $P_2(\mathbb{R}) \Rightarrow T$ is diagonalizable.



(iii) $T(p(x)) = p(0) + p(1)(x + x^2)$

$$T(1) = 1 + x + x^2$$

$$T(x) = x + x^2$$

$$T(x^2) = (x + x^2)$$

$$[T]_{\beta}^{\beta} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\text{Char. poly of } T = \det \begin{pmatrix} 1-\lambda & 0 & 0 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda) \left((1-\lambda)^2 - 1 \right) = (1-\lambda)(\lambda^2 - 2\lambda)$$

$$= \lambda(1-\lambda)(\lambda-2)$$

Hence the eigenvalues of T are $0, 1, 2$.

Since T has 3 distinct eigenvalues, we have
 T is diagonalizable. _____ .

Problem 2: Let T be an invertible linear operator on a finite-dimensional vector space V . Prove that the eigenspace of T corresponding to λ is the same as the eigenspace of T^{-1} corresponding to λ^{-1} . Furthermore, prove that if T is diagonalizable, then so is T^{-1} .

Proof: Since T is invertible, 0 is not an eigenvalue of T .

Let λ be an eigenvalue of T and v be a non-zero vector s.t.

$$Tv = \lambda v$$

$$T^{-1}T = I$$

$$T^{-1}Tv = v$$

$$\Rightarrow T^{-1}(\lambda v) = v$$

$$\Rightarrow T^{-1}v = \left(\frac{1}{\lambda}\right)v.$$

$\Rightarrow v$ is an eigenvector of T^{-1} corresponding to $1/\lambda$.

Hence

Eigenspace of T corresponding to $\lambda \subseteq$ Eigenspace of T^{-1} corresponding to $1/\lambda$.

||^{ly} Eigenspace of T^{-1} corresp. to $1/\lambda \subseteq$ Eigenspace of T corresp. to λ .

$\Rightarrow$ the eigenspaces are equal.

Since T is diagonalizable, $\exists$ a basis $\beta = (v_1, \dots, v_n)$
of eigenvectors of T . $\Rightarrow$

$\exists$ a basis of T^{-1} consisting of eigenvectors.

$\Rightarrow T^{-1}$ - diagonalizable. ■

Problem 3: Let V be a vector space and T be a linear operator on V . Suppose W is the T -cyclic subspace of V generated by a non-zero vector v . Prove that for every $w \in W$, $\exists$ a polynomial $g(t)$ s.t. $g(T)v = w$. Moreover, if W is finite dimensional, then g can be picked such that $\deg(g) \leq \dim(W)$.

Proof: $W = \text{span} \{ v, Tv, T^2v, \dots \}$

Let $w \in W$

Let $a_1, \dots, a_n$ be s.t.

$$\begin{aligned} w &= a_1 T^{k_1} v + a_2 T^{k_2} v + \dots + a_n T^{k_n} v \\ &= (a_1 T^{k_1} + a_2 T^{k_2} + \dots + a_n T^{k_n}) v \end{aligned}$$

Let $g(t) = a_1 t^{k_1} + a_2 t^{k_2} + \dots + a_n t^{k_n}.$

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Suppose $\dim(W) = m+1$

Claim: $T^k v \in \text{span}\{v, Tv, \dots, T^m v\} \quad \forall k \geq m+1.$

$$\Rightarrow W = \text{span}\{v, Tv, \dots, T^m v\}.$$

$$\Rightarrow w \in \text{span}\{v, Tv, \dots, T^m v\}.$$

$$\Rightarrow \exists b_0, \dots, b_m \text{ s.t.}$$

$$w = b_0 v + b_1 Tv + \dots + b_m T^m v$$

$$\text{define } g(t) = b_0 + b_1 t + \dots + b_m t^m$$

The $\deg(g(t)) \leq m$
 & $g(T)v = w$ _____ $\blacksquare$.

Problem 4: Let $A \in M_n(\mathbb{R})$. Then prove that
 $\dim(W) \leq n$ where $W = \text{span}\{I, A, A^2, \dots\}$

Proof: Since $A \in M_n(\mathbb{R})$, consider

$$L_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$\Rightarrow$ Char. poly of $L_A =$ char poly of A has

$$\deg = n.$$

$\Rightarrow f(\lambda)$ Char. poly of A has deg n .

$$f(\lambda) = (-1)^n \lambda^n + \dots + a_0$$

The Cayley-Hamilton theorem

$$f(L_A) \equiv 0$$

$$\Rightarrow (-1)^n L_A^n + \dots + a_0 I \equiv 0$$

$$\Rightarrow (-1)^n A^n + a_{n-1} A^{n-1} + \dots + a_0 I \equiv 0.$$

$$\Rightarrow A^n \in \text{span}(I, A, A^2, \dots, A^{n-1}).$$

$$\Rightarrow A^k \in \text{span}(I, A, A^2, \dots, A^{n-1}) \quad \forall k \geq n.$$

$$\Rightarrow W \subseteq \text{span}(I, A, A^2, \dots, A^{n-1})$$

$$\text{Hence } W = \text{span}(I, A, A^2, \dots, A^{n-1})$$

Since W has a spanning set of size n ,
we can conclude that

$$\dim(W) \leq n.$$



Problem 5: Let V be an inner product space &

$u, v \in V$. Then prove that $\langle u, v \rangle = 0$ iff

$$\|u\| \leq \|u + av\| \quad \forall a \in \mathbb{F}.$$

Proof:

$$\begin{aligned} & \text{If } \langle u, v \rangle = 0, \\ & \|u + av\|^2 = \|u\|^2 + |a|^2 \|v\|^2 \quad \forall a \in \mathbb{F} \\ & \quad \quad \quad (> 0) \end{aligned}$$

$$\Rightarrow \|u\|^2 \leq \|u + av\|^2 \quad \forall a \in \mathbb{F}$$

$$\Rightarrow \|u\| \leq \|u + av\| \quad \forall a \in \mathbb{F}.$$

Conversely, let $\|u\| \leq \|u + av\| \quad \forall a \in \mathbb{F}$.

$$\begin{aligned}\|u + av\|^2 &= \langle u + av, u + av \rangle \\ &= \|u\|^2 + a \langle v, u \rangle + \bar{a} \langle u, v \rangle + |a|^2 \|v\|^2\end{aligned}$$

$$\|u + av\|^2 - \|u\|^2 = 2 \operatorname{Re}(\bar{a} \langle u, v \rangle) + |a|^2 \|v\|^2$$

$$\|u + av\|^2 - \|u\|^2 \geq 0 \quad \forall a \in \mathbb{F}$$

$$\Rightarrow 2 \operatorname{Re}(\bar{a} \langle u, v \rangle) + |a|^2 \|v\|^2 \geq 0 \quad \longrightarrow (*)$$

If $v = 0$, the proof follows.

When $v \neq 0$, $a = -\frac{\langle u, v \rangle}{\|v\|^2}$

Then L.H.S of (*) will be

$$\begin{aligned} & 2 \operatorname{Re} \left(-\frac{\overline{\langle u, v \rangle} \langle u, v \rangle}{\|v\|^2} \right) + \frac{|\langle u, v \rangle|^2 \|v\|^2}{\|v\|^4} \\ &= 2 \operatorname{Re} \left(-\frac{|\langle u, v \rangle|^2}{\|v\|^2} \right) + \frac{|\langle u, v \rangle|^2}{\|v\|^2} \\ &= -\frac{2 |\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2} \end{aligned}$$

$$= - \frac{2 |\langle u, v \rangle|}{\|v\|^2}$$

$$B_y \quad (*) \quad - \frac{2 |\langle u, v \rangle|}{\|v\|^2} \geq 0$$

$$\Rightarrow - |\langle u, v \rangle| \geq 0$$

$$\Rightarrow |\langle u, v \rangle| \leq 0$$

$$\Rightarrow |\langle u, v \rangle| = 0$$

$$\Rightarrow \langle u, v \rangle = 0 \quad \underline{\hspace{1cm}}$$



Problem 6:

Prove that

$$\left(\sum_{j=1}^n a_j b_j \right)^2 \leq \left(\sum_{j=1}^n j a_j^2 \right) \left(\sum_{j=1}^n \frac{b_j^2}{j} \right).$$

$\forall a_i, b_i \in \mathbb{R}$ where $i = 1, 2, \dots, n$.

Proof: Consider $\mathbb{R}^n$ with the standard inner product.

Let $(a_1, \dots, a_n), (b_1, \dots, b_n) \in \mathbb{R}^n$.

By Cauchy-Schwarz inequality, we have

$$| \langle (a_1, \dots, a_n), (b_1, \dots, b_n) \rangle | \leq \| (a_1, \dots, a_n) \| \| (b_1, \dots, b_n) \|$$

Define

$$a'_i = \sqrt{i} a_i$$

by

$$b'_i = b_i / \sqrt{i}$$

Apply C-S to $(a'_1, \dots, a'_n)$ and $(b'_1, \dots, b'_n)$

$$| \langle (a'_1, \dots, a'_n), (b'_1, \dots, b'_n) \rangle |^2 \leq \| (a'_1, \dots, a'_n) \|^2 \| (b'_1, \dots, b'_n) \|^2$$

$$\Rightarrow \left| (a'_1 b'_1 + a'_2 b'_2 + \dots + a'_n b'_n) \right|^2 \leq (a_1'^2 + \dots + a_n'^2) (b_1'^2 + \dots + b_n'^2)$$

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$$\Rightarrow (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq ((1a_1)^2 + (\sqrt{2}a_2)^2 + \dots + (\sqrt{n}a_n)^2) \left(\frac{b_1^2}{1} + \frac{b_2^2}{2} + \dots + \frac{b_n^2}{n} \right)$$

$$\leq \left(\sum_{j=1}^n j a_j^2 \right) \left(\sum_{j=1}^n \frac{b_j^2}{j} \right) \quad \square$$

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Problem 7: Let V be a complex inner product space. Prove that

$$\langle u, v \rangle = \frac{(\|u+v\|^2 - \|u-v\|^2) + i(\|u+iv\|^2 - \|u-iv\|^2)}{4}$$

$$\begin{aligned} & (\|u+v\|^2 - \|u-v\|^2) + i(\|u+iv\|^2 - \|u-iv\|^2) = \\ & (\langle u+v, u+v \rangle - \langle u-v, u-v \rangle) + i(\langle u+iv, u+iv \rangle - \langle u-iv, u-iv \rangle) \\ & = (\|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle) - (\|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle) \\ & \quad + i(\langle u, u \rangle + i\bar{\langle u, v \rangle} + i\langle v, u \rangle + i\bar{\langle v, v \rangle}) - \end{aligned}$$

$$(\langle u, u \rangle - i \langle v, u \rangle - i \langle u, v \rangle + i i \langle v, v \rangle)$$

$$= 2 (\langle u, v \rangle + \langle v, u \rangle) + i (\|u\|^2 - i \langle u, v \rangle + i \langle v, u \rangle + \|v\|^2) - (\|u\|^2 - i \langle v, u \rangle + i \langle u, v \rangle + \|v\|^2)$$

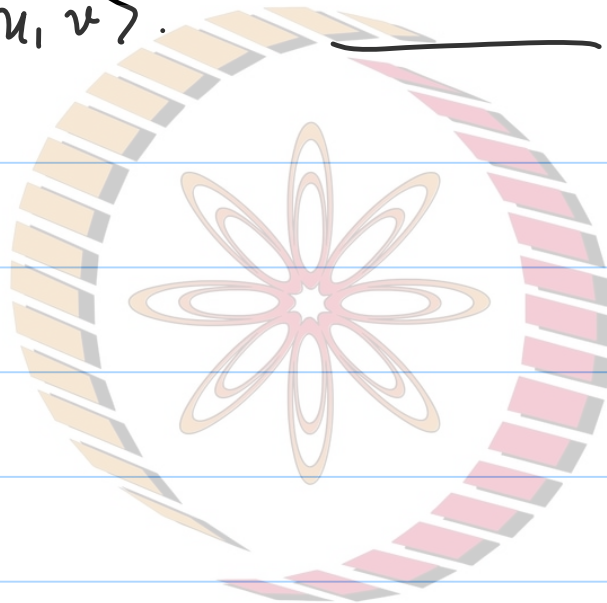
$$= 2 (2 \operatorname{Re}(\langle u, v \rangle) + i 2 (i (\underbrace{\langle v, u \rangle}_{\overline{\langle u, v \rangle}} \underbrace{=}_{a+ib} - \underbrace{\langle u, v \rangle}_{a-ib}))$$

$$= 4 \operatorname{Re}(\langle u, v \rangle) + i^2 2 (2i \operatorname{Im}(\langle v, u \rangle))$$

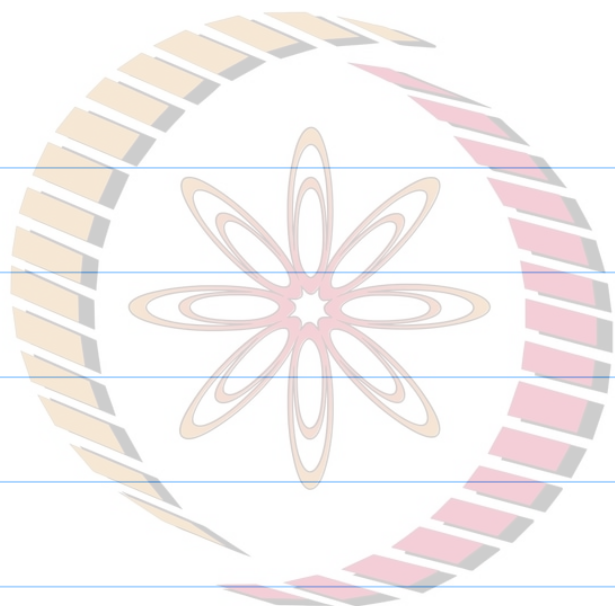
$$= 4 (\operatorname{Re}(\langle u, v \rangle) - i \operatorname{Im}(\overline{\langle u, v \rangle})).$$

$$= 4 (\operatorname{Re}(\langle u, v \rangle) + i \operatorname{Im}(\langle u, v \rangle)).$$

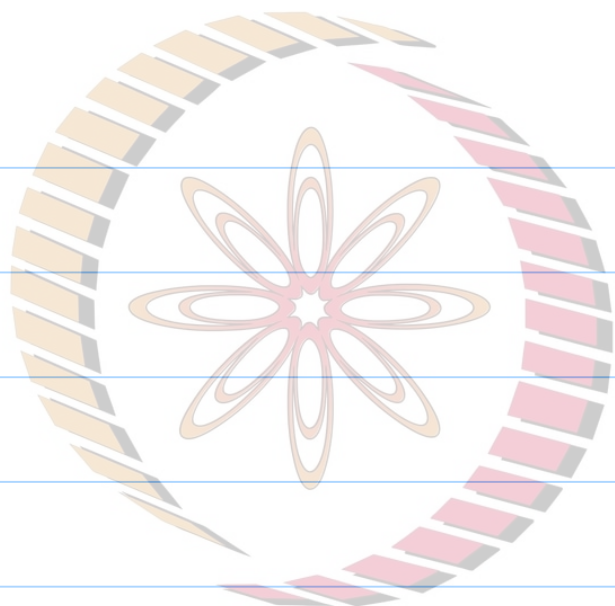
$$= 4 \langle u, v \rangle.$$



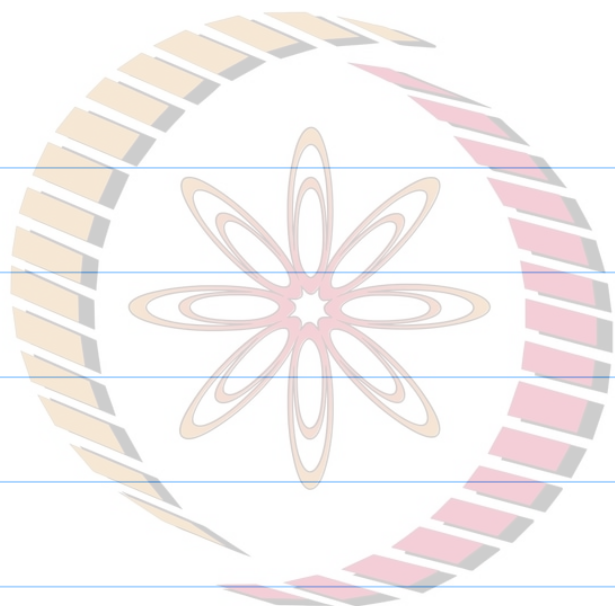
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